Westmere University
Faculty of Earth and Environmental Sciences
Department of Water Resources Engineering

Introduction to Hydrology

HYD 201 · Course syllabus · Autumn semester 2026

Credits	3 credit hours
Lectures	Tuesday and Thursday, 10:00–11:20, Room G.14
Laboratory	Wednesday, 14:00–16:50, Hydraulics Laboratory
Instructor	Dr Rosalind Achterberg
Office hours	Monday 13:00–15:00, or by appointment
Email	r.achterberg@westmere.example

Course description

This course introduces the physical processes that move water through the landscape, from precipitation and interception to infiltration, groundwater recharge and streamflow. Students learn to measure these processes in the field and to describe them with simple quantitative models. Weekly laboratory sessions use real data from the university's experimental catchment on the Ashby Brook.

The course is a prerequisite for HYD 305 Groundwater Hydraulics and for the water resources specialisation. Prior completion of MATH 120 or an equivalent course in calculus is required.

Learning outcomes

By the end of the course, students will be able to:

- describe the components of the hydrological cycle and estimate a catchment water balance;
- analyse rainfall records and construct intensity–duration–frequency curves;
- measure streamflow with the velocity–area method and build a stage–discharge rating curve;
- apply the rational method and the unit hydrograph to estimate design floods;
- write a clear technical report that states its assumptions and uncertainties.

Weekly schedule

Week	Dates	Topic and laboratory	Reading
1	1–5 Sep	The hydrological cycle and catchments Lab: Catchment delineation from maps	Ch. 1
2	8–12 Sep	Precipitation: measurement and spatial averaging Lab: Thiessen polygons and isohyets	Ch. 2
3	15–19 Sep	Rainfall statistics and design storms Lab: IDF curves from gauge records	Ch. 3
4	22–26 Sep	Interception and evaporation Lab: Pan evaporation data	Ch. 4

Week	Dates	Topic and laboratory	Reading
5	29 Sep–3 Oct	Evapotranspiration and the Penman equation Lab: Reference ET from weather data	Ch. 4
6	6–10 Oct	Infiltration and soil moisture Field: Double-ring infiltrometer tests	Ch. 5
7	13–17 Oct	Groundwater and recharge Lab: Well hydrographs	Ch. 6
8	20–24 Oct	Mid-term examination Lab: Review session	—
9	27–31 Oct	Streamflow measurement Field: Velocity–area gauging on the Ashby Brook	Ch. 7
10	3–7 Nov	Rating curves and flow records Lab: Building a rating curve	Ch. 7
11	10–14 Nov	Hydrograph analysis and baseflow separation Lab: Storm hydrograph analysis	Ch. 8
12	17–21 Nov	Unit hydrograph theory Lab: Deriving a unit hydrograph	Ch. 8
13	24–28 Nov	Flood frequency analysis Lab: Annual maximum series	Ch. 9
14	1–5 Dec	Design floods and course review Lab: Rational method exercise	Ch. 10

Assessment

Component	Weight	Due
Laboratory reports (five)	25%	Weeks 3 to 12
Field-trip report	15%	20 November
Mid-term examination	20%	22 October
Final examination	40%	December examination period
Total	**100%**	

Grades follow the university's standard scale, and a mark of 50% overall is required to pass the course.

Course policies

Attendance

Attendance at laboratory and field sessions is compulsory because the data collected there are used in the reports. Students who miss a session for a documented reason will be given an equivalent data set.

Late submissions

(a) Reports submitted up to 24 hours late lose 10% of the available marks.

(b) Reports submitted between 24 hours and one week late lose 30%.

(c) Reports more than one week late are not marked unless an extension was agreed in advance.

Academic integrity

You may discuss laboratory data with other students, but every report must be written individually. Generative tools may be used for spelling and grammar only, and their use must be declared[1].

Suspected misconduct is referred to the faculty's academic integrity panel.

[1] See the faculty guidance on the declaration of digital tools, section 4.2.